

Lecture 2: Double Descent in Linear Regression

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1 Linear Models and Their Extensions

Linear model A *linear model* is the simplest predictive model, mapping an input vector $\mathbf{x} \in \mathbb{R}^p$ to a scalar output $y \in \mathbb{R}$ using a linear function:

$$y = f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}, \quad (1)$$

where:

- $\mathbf{x} \in \mathbb{R}^p$: the input vector with p features,
- $\mathbf{w} \in \mathbb{R}^p$: the parameter (weight) vector,
- $y \in \mathbb{R}$: the predicted scalar output.

Linear regression Given training data

$$\{(\mathbf{x}_i, y_i)\}_{i=1}^n, \quad \mathbf{x}_i \in \mathbb{R}^p, y_i \in \mathbb{R},$$

we construct the data matrix

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} \in \mathbb{R}^{n \times p}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n.$$

Linear regression learns the parameter vector $\mathbf{w}$ by minimizing the squared loss:

$$\mathcal{L}(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2. \quad (2)$$

The system $\mathbf{X}\mathbf{w} = \mathbf{y}$ has a *unique* solution if and only if the following conditions hold:

1. **Full column rank:**

$$\text{rank}(X) = p.$$

This ensures that the p columns of X are linearly independent.

2. **Sufficient number of equations:**

$$n \geq p.$$

If $n < p$, the system is underdetermined and cannot have a unique solution.

3. **Consistency:**

$$y \in \text{Col}(X).$$

That is, the vector y must lie in the column space of X .

If $\mathbf{X}^\top \mathbf{X}$ is invertible (when the 3 above conditions are satisfied), the minimizer is given by the equation:

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (3)$$

Derivation of the optimal solution We can rewrite the loss as

$$\mathcal{L}(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 = (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w}).$$

Expanding the quadratic form gives

$$\mathcal{L}(\mathbf{w}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\mathbf{w} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w}.$$

Taking the gradient with respect to $\mathbf{w}$ and using standard matrix calculus rules,

$$\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}) = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\mathbf{w}.$$

Setting the gradient to zero to find a minimizer,

$$\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}) = 0 \implies -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\mathbf{w} = 0,$$

which simplifies to the *normal equation*

$$\mathbf{X}^\top \mathbf{X}\mathbf{w} = \mathbf{X}^\top \mathbf{y}.$$

If $\mathbf{X}^\top \mathbf{X}$ is invertible, this linear system has the unique solution

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y},$$

which is precisely Eq. (3).

If $\mathbf{X}^\top \mathbf{X}$ is not invertible (e.g., when the columns of $\mathbf{X}$ are linearly dependent or $p > n$), the normal equation

$$\mathbf{X}^\top \mathbf{X}\mathbf{w} = \mathbf{X}^\top \mathbf{y}$$

may have infinitely many solutions. In this case, a canonical choice is the *minimum-norm* least-squares solution given by the Moore–Penrose pseudoinverse:

$$\hat{\mathbf{w}} = \mathbf{X}^+ \mathbf{y},$$

where $\mathbf{X}^+$ denotes the pseudoinverse of $\mathbf{X}$.

Random feature model A random feature model is simply a linear model applied to a nonlinear feature map, where the nonlinear features are generated by fixed random projections. Formally, random feature models are defined as:

$$\hat{\mathbf{y}} = \sigma(\mathbf{X})\mathbf{w}, \quad \text{where } \mathbf{X} \in \mathbb{R}^{p \times n}, \mathbf{w} \in \mathbb{R}^m,$$

where

$$\sigma(\mathbf{X}) = \phi(\mathbf{W}_0 \mathbf{X}), \quad \mathbf{W}_0 \sim \mathcal{N}(0, \mathbf{I})$$

The parameter vector $\mathbf{w}$ is learned by optimizing the following loss function:

$$\mathcal{L}(\mathbf{w}) = \|\mathbf{y} - \sigma(\mathbf{X})\mathbf{w}\|^2,$$

where $\mathbf{y} \in \mathbb{R}^n$ are ground-truth labels.

This model mixes random nonlinearities (via ϕ) with linear regression over random projections.

Neural network A neural network defines a function $f: \mathbb{R}^p \rightarrow \mathbb{R}$ that maps an input vector $\mathbf{x} \in \mathbb{R}^p$ to a scalar output $y \in \mathbb{R}$:

$$y = f(\mathbf{x}) = \mathbf{W}_2 \sigma(\mathbf{W}_1 \mathbf{x}), \tag{4}$$

where:

- $\mathbf{x} \in \mathbb{R}^p$: the input vector with p features.
- $\mathbf{W}_1 \in \mathbb{R}^{h \times p}$: the weight matrix of the first (input-to-hidden) layer, mapping from the p -dimensional input space to a hidden layer of size h .
- $\sigma: \mathbb{R}^h \rightarrow \mathbb{R}^h$: a nonlinear activation function (e.g., ReLU or sigmoid) applied elementwise to the hidden layer output.
- $\mathbf{W}_2 \in \mathbb{R}^{1 \times h}$: the weight vector of the second (hidden-to-output) layer, mapping the hidden layer activations to a scalar output.
- $y \in \mathbb{R}$: the final output of the network, which is one scalar for a given input vector.

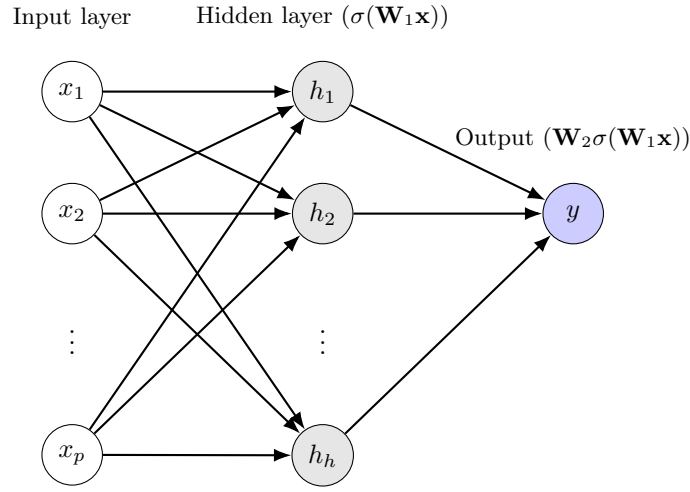


Figure 1: A graphical representation of a neural network with two layers.

Eq. 4 represents a simple *two-layer (single hidden layer)* feedforward neural network with no bias terms. A graphical representation is shown in Figure 1.

In the following, we will study a simpler linear model that can be seen as a first-order approximation of a neural network around its initialization. Another related model is the random feature model discussed below.

2 Overdetermined Case

We consider a linear model in the *underparameterized regime*, where the number of samples exceeds the number of parameters, i.e., $n > p$. The vector of outputs is given by:

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \mathbf{n} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n), \quad \mathbf{X} \in \mathbb{R}^{n \times p}, \quad \mathbf{w} \in \mathbb{R}^p,$$

where $\mathbf{X}$ is the input data matrix, $\mathbf{w}$ is the true weight vector, and $\mathbf{n}$ is Gaussian noise.

To learn an estimate $\hat{\mathbf{w}}$ of the true weights, we minimize the squared loss:

$$\mathcal{L}(\hat{\mathbf{w}}) = \|\mathbf{y} - \mathbf{X}\hat{\mathbf{w}}\|^2.$$

Since this is a linear least-squares problem and $\mathbf{X}^\top \mathbf{X}$ is invertible (due to $n > p$), the minimizer has a closed-form solution:

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Proposition 1. Let $\gamma = \frac{p}{n}$ denote the aspect ratio. Then the expected squared error of the estimator satisfies:

$$\mathbb{E} [\|\mathbf{w} - \hat{\mathbf{w}}\|^2] = \sigma^2 \cdot \frac{\gamma}{1 - \gamma}$$

Proof. We have

$$\begin{aligned} \|\mathbf{w} - \hat{\mathbf{w}}\|^2 &= \|\mathbf{w} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}\|^2 \\ &= \|\mathbf{w} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\mathbf{w} + \mathbf{n})\|^2 \\ &= \|\mathbf{w} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\mathbf{w} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{n}\|^2 \\ &= \|\mathbf{w} - \mathbf{w} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{n}\|^2 \\ &= \|(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{n}\|^2 \\ &= \mathbf{n}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-2} \mathbf{X}^\top \mathbf{n}. \end{aligned}$$

In the following, we denote by $\text{Tr}(\mathbf{A})$ the trace of a matrix $\mathbf{A}$. Since $\mathbf{n} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$, we obtain:

$$\begin{aligned} \mathbb{E}_{\mathbf{n}} \|\mathbf{w} - \hat{\mathbf{w}}\|^2 &= \mathbb{E}_{\mathbf{n}} [\mathbf{n}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-2} \mathbf{X}^\top \mathbf{n}] \\ &= \sigma^2 \text{Tr} [\mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-2} \mathbf{X}^\top] \\ &= \sigma^2 \text{Tr} [(\mathbf{X}^\top \mathbf{X})^{-1}] \\ &= \sigma^2 \cdot \frac{p}{n} \text{tr} \left[\left(\frac{1}{n} \mathbf{X}^\top \mathbf{X} \right)^{-1} \right], \end{aligned}$$

where $\text{tr} = \frac{1}{p} \text{Tr}$ is the normalized trace (note that $\mathbf{X}^\top \mathbf{X}$ is an $p \times p$ matrix).

Assume $\mathbf{X}$ is a standard Gaussian random matrix. Then $\hat{\Sigma} = \frac{1}{n} \mathbf{X}^\top \mathbf{X}$ is a Wishart matrix, and as discussed earlier, we have

$$S_n(0) = \mathbb{E} \left[\text{tr} \left(\hat{\Sigma} \right)^{-1} \right].$$

Therefore, we conclude that

$$\mathbb{E}_{\mathbf{n}, \mathbf{X}} \|\mathbf{w} - \hat{\mathbf{w}}\|^2 \xrightarrow[n \rightarrow \infty]{p=\gamma n} \sigma^2 \cdot \gamma \cdot S(0)$$

where

$$S(z) = \int \frac{1}{t-z} \psi_{\text{MP}}(t) dt \quad \text{is the Stieltjes transform}$$

Solving the Marchenko–Pastur equation for $z = 0$:

$$1 + \gamma S(0) = \frac{S(0)}{1 + \gamma S(0)} \quad \Rightarrow \quad 1 = \frac{S(0)}{1 + \gamma S(0)} \quad \Rightarrow \quad S(0) = \frac{1}{1 - \gamma}$$

Therefore:

$$\mathbb{E} [\|\mathbf{w} - \hat{\mathbf{w}}\|^2] = \sigma^2 \cdot \frac{\gamma}{1 - \gamma}$$

□

An illustration of the error is shown in Figure 2 where it can be seen that the error *explodes* when $\gamma = 1$, i.e., $p = n$.

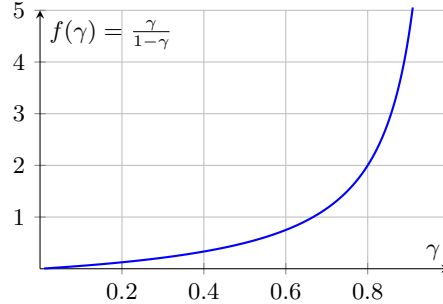


Figure 2: Illustration of the function $\sigma^2 \cdot \frac{\gamma}{1 - \gamma}$ for $\sigma^2 = \frac{1}{2}$.

3 Underdetermined Case: $n < p$

In this case, the system has infinitely many solutions:

$$\mathbf{y} = \mathbf{X} \hat{\mathbf{w}} \quad \Rightarrow \quad \text{infinitely many } \hat{\mathbf{w}}$$

We will therefore choose the following solution, which is the one with the minimal norm as shown in Lemma 2:

$$\hat{\mathbf{w}} = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{y} \quad \text{if } \text{rank}(\mathbf{X}) = n$$

Lemma 2. Let $\mathbf{A} \in \mathbb{R}^{n \times p}$ with $n < p$ and consider the system:

$$\mathbf{A} \mathbf{x} = \mathbf{y}, \quad \mathbf{x} \in \mathbb{R}^{p \times 1}, \quad \mathbf{y} \in \mathbb{R}^{n \times 1}$$

Assume $\mathbf{A}$ has full rank: $\text{rank}(\mathbf{A}) = n$. Then

$$\mathbf{x}_0 = \mathbf{A}^\top (\mathbf{A} \mathbf{A}^\top)^{-1} \mathbf{y}$$

is a solution of $\mathbf{A} \mathbf{x} = \mathbf{y}$ and has the smallest norm among all solutions.

Proof. 1. Verify that $\mathbf{x}_0$ is a solution:

$$\mathbf{A}\mathbf{x}_0 = \mathbf{A}\mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}\mathbf{y} = \mathbf{y}$$

2. Show minimality: Assume $\mathbf{x}$ is any solution: $\mathbf{A}\mathbf{x} = \mathbf{y}$. Then:

$$\langle \mathbf{x} - \mathbf{x}_0, \mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}\mathbf{y} \rangle = \langle \mathbf{A}(\mathbf{x} - \mathbf{x}_0), (\mathbf{A}\mathbf{A}^\top)^{-1}\mathbf{y} \rangle = 0 \quad \text{since } \mathbf{A}(\mathbf{x} - \mathbf{x}_0) = 0$$

Therefore:

$$\|\mathbf{x}\|^2 = \|\mathbf{x} - \mathbf{x}_0 + \mathbf{x}_0\|^2 = \|\mathbf{x} - \mathbf{x}_0\|^2 + \|\mathbf{x}_0\|^2 \geq \|\mathbf{x}_0\|^2$$

Hence, $\mathbf{x}_0$ is the minimum norm solution. □

Error term Next, we characterize the error term between the true and predicted solutions.

Proposition 3. Let $\gamma = \frac{p}{n}$ denote the aspect ratio. Then the expected squared error of the estimator satisfies:

$$\mathbb{E} [\|\mathbf{w} - \hat{\mathbf{w}}\|^2] = \begin{cases} \sigma^2 \cdot \frac{\gamma}{1-\gamma}, & \gamma < 1 \\ \sigma^2 \cdot \frac{1}{1-\gamma} + \|\mathbf{w}\|^2 \left(1 - \frac{1}{\gamma}\right), & \gamma > 1 \end{cases}$$

Proof. Recall that the minimal norm solution is:

$$\hat{\mathbf{w}} = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{y}$$

We want to compute:

$$\|\mathbf{w} - \hat{\mathbf{w}}\|^2 = \|\mathbf{w} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{y}\|^2$$

Assuming that $\mathbf{y} = \mathbf{X}\mathbf{w} + \mathbf{n}$, we get:

$$\begin{aligned} \|\mathbf{w} - \hat{\mathbf{w}}\|^2 &= \|\mathbf{w} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{y}\|^2 \\ &= \|\mathbf{w} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}(\mathbf{X}\mathbf{w} + \mathbf{n})\|^2 \\ &= \|\mathbf{w} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{w} + \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n}\|^2 \\ &= \|(\mathbf{I} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X})\mathbf{w} + \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n}\|^2. \end{aligned}$$

Let:

$$\mathbf{w}_1 = (\mathbf{I} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X})\mathbf{w}, \quad \mathbf{w}_2 = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n}.$$

Note $\mathbf{w}_1 \perp \mathbf{w}_2$ because:

$$\begin{aligned} \mathbf{w}_1^\top \mathbf{w}_2 &= \mathbf{w}^\top (\mathbf{I} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X})\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n} \\ &= \mathbf{w}^\top (\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n}) \\ &= \mathbf{w}^\top (\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n}) = 0. \end{aligned}$$

So:

$$\|\mathbf{w} - \hat{\mathbf{w}}\|^2 = \underbrace{\|\mathbf{w}_1\|^2}_{\text{Bias term}} + \underbrace{\|\mathbf{w}_2\|^2}_{\text{Variance term}}.$$

Variance term: Consider the variance term first:

$$\begin{aligned} \mathbb{E}_{\mathbf{n}} \|\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n}\|^2 &= \mathbb{E}_{\mathbf{n}} [\mathbf{n}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n}] \\ &= \sigma^2 \text{Tr}((\mathbf{X}\mathbf{X}^\top)^{-1}) \\ &= \frac{\sigma^2 n}{p} \text{tr} \left[\left(\frac{1}{p} \mathbf{X}\mathbf{X}^\top \right)^{-1} \right], \end{aligned}$$

where $\text{tr} = \frac{1}{n} \text{Tr}$ is the normalized trace (note that $\mathbf{X}\mathbf{X}^\top$ is an $n \times n$ matrix).

Note that we obtain the same result as the overdetermined case, but now $\mathbf{X}^\top\mathbf{X}$ is replaced by $\mathbf{X}\mathbf{X}^\top$. This means that the variables n and p are exchanged and the same derivation shows that γ is replaced by $\frac{1}{\gamma}$. Therefore, we conclude that

$$\mathbb{E}_{\mathbf{n}} \|\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{n}\|^2 = \sigma^2 \cdot \frac{1/\gamma}{1 - \sqrt{1/\gamma}} = \sigma^2 \cdot \frac{1}{\gamma - 1}$$

Bias term:

$$\begin{aligned}
 \|\mathbf{w} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{w}\|^2 &= (\mathbf{w}^\top - \mathbf{w}^\top \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X})(\mathbf{w} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{w}) \\
 &= \mathbf{w}^\top \mathbf{w} - \mathbf{w}^\top \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{w} - \mathbf{w}^\top \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{w} + \mathbf{w}^\top \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{w} \\
 &= \mathbf{w}^\top \mathbf{w} - \mathbf{w}^\top \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{w} \\
 &= \|\mathbf{w}\|^2 \left(1 - \frac{\mathbf{w}^\top}{\|\mathbf{w}\|} \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X} \frac{\mathbf{w}}{\|\mathbf{w}\|}\right).
 \end{aligned}$$

Let $\mathbf{v} = \frac{\mathbf{w}}{\|\mathbf{w}\|}$. Due to the rotational invariance of the Gaussian distribution, we have:

$$\mathbf{v}^\top \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{v} = \frac{1}{p} \text{Tr} [\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}] = \frac{1}{p} \text{Tr}(\mathbf{I}_n) = \frac{n}{p}$$

We conclude that

$$\|\mathbf{w} - \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}\mathbf{w}\|^2 = \|\mathbf{w}\|^2 \left(1 - \frac{n}{p}\right) = \|\mathbf{w}\|^2 \left(1 - \frac{1}{\gamma}\right)$$

Bias + Variance: Putting things together, we obtain the following expression for the error:

$$\mathbb{E} [\|\mathbf{w} - \hat{\mathbf{w}}\|^2] = \begin{cases} \sigma^2 \cdot \frac{\gamma}{1-\gamma}, & \gamma < 1 \\ \sigma^2 \cdot \frac{1}{\gamma-1} + \|\mathbf{w}\|^2 \left(1 - \frac{1}{\gamma}\right), & \gamma > 1 \end{cases}$$

□

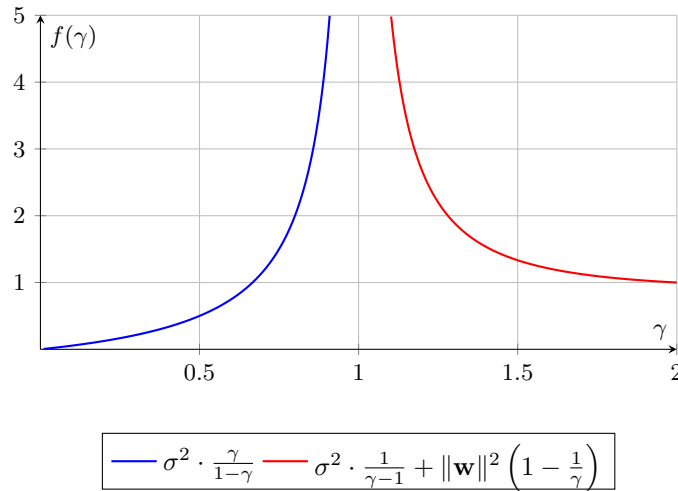


Figure 3: Illustration of the error term with $\sigma^2 = \frac{1}{2}$ and $\|\mathbf{w}\|^2 = 1$. We see that the error explodes at $\gamma = 1$.